

INPUTS

Procedural binary axis mask

Grayscale SEM image

A Enabling example — a binary mask from real SEM

CycleGAN

SEM-like synthetic
images

U-Net

SEM-derived
axis mask

B PLECTA — overlap-aware instance segmentation (this paper)

OR

EVALUATED: mask in → instances out

Binary filament-axis mask

STEP 1

Skeleton graph:
arms, stubs,
junction nodes

STEP 2

Frame at
each stub

STEP 3

Exact matching
at each junction

STEP 4

Gated bridging
of mask gaps

STEP 5

Paint instance
layers

8 rounds

Overlap-aware instance layers
(a crossing pixel has two owners)

Optional: width measurement
and ribbon rendering

Rendered ribbon
instances

default off — changes the representation, not the identities

Legend: Learned component PLECTA geometric step Needs the SEM image Data artefact